

Titanic Dataset EDA Report

1. Dataset Overview

- Total entries: 891
- Features: PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
- Missing Data: Age, Cabin, Embarked

2. Survival Overview

- 0 = Did not survive, 1 = Survived
- Observation: Most passengers did not survive (as shown in the bar chart).

3. Gender vs Survival

- Observation: Females had a higher survival rate than males.

4. Class vs Survival

- Observation: 1st class passengers were more likely to survive.

5. Age Distribution

- Observation: Wide distribution with many young adults and fewer seniors.

6. Age by Class

- Observation: Younger passengers more common in 3rd class.

7. Heatmap - Missing Data

- Observation: Cabin has the most missing values. Age has some missing entries.

8. Pairplot Analysis

- Observation: Higher fares and lower Pclass (1st class) correlate with higher survival.

9. Correlation Heatmap

- Highest correlation with survival: Sex (-0.54), Pclass (-0.34), Fare (0.26)

Summary of Findings

- Gender and Passenger Class were strong indicators of survival.
- Fare and Age had a weaker but visible relationship with survival.
- Cabin is too incomplete to be useful without imputation.

Next Steps

- Handle missing values (e.g., impute Age).
- Engineer features (e.g., family size from SibSp + Parch).
- Use this EDA to guide modeling and prediction.